
PH102: Physics-2

Electrodynamics

Lecture # 09

Electrostatics – Special techniques:
The method of images and
Separation of variables

07.02.2024

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Previous lecture

- Any solution of Laplace's equation which satisfies the same boundary conditions must be the only solution regardless of the method used. This is known as the *uniqueness theorem*.
- The theorem applies to any solution of Poisson's or Laplace's equation in a given region or closed surface.

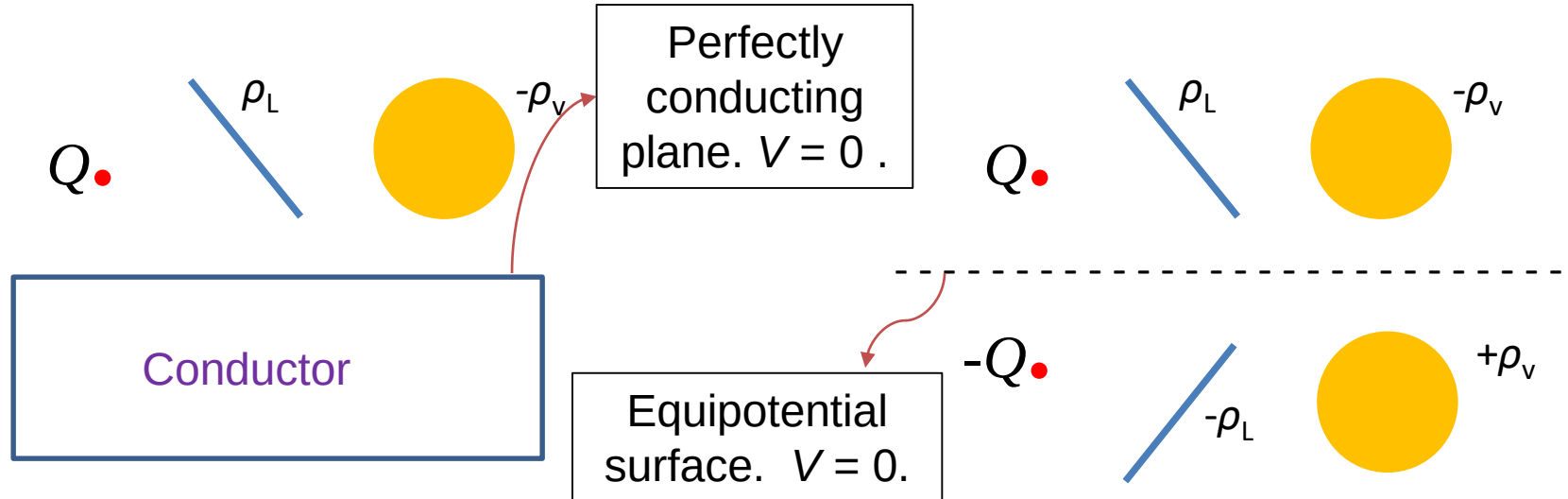
Method of images

- The primary task of **method of images** is to determine V , $\mathbf{E}$, $\mathbf{D}$ and ρ due to charges **in the presence of conductors**.
- By this method we **avoid solving Poisson's or Laplace's equation** but rather **utilize the fact that a conducting surface is an equipotential**.
- Although the method does not apply to all electrostatic problems, it can reduce a formidable problem to a simple one.
- The image theory states that **a given charge configuration above an infinite grounded perfect conducting plane may be replaced by the charge configuration itself, its image, and an equipotential surface in place of the conducting plane**.

The basic idea is that a charge (point charge or charge distribution) reflects in a conductor much the same as light reflects in a mirror.

In applying the image method, two conditions must always be satisfied.

- (1) The image charges must be located in the conducting region.
- (2) The image charge must be located such that on the conducting surface the potential is zero or constant.



The method of images solves for the fields in a given problem by replacing it with an equivalent problem, which is easier to solve and which has the same solution in the required domain or for which the solution is known or can be obtained readily.

Classic image problem: The infinite grounded conducting plane

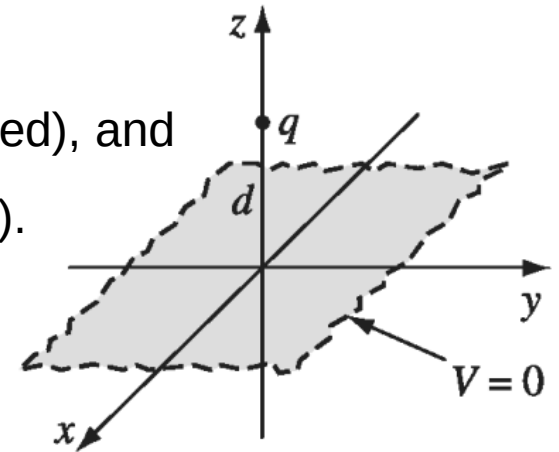
Suppose a point charge q is held a distance d above an infinite grounded conducting plane. What is the potential in the region above the plane?

Our problem is to solve Poisson's equation in the region $z > 0$, with a single point charge q at $(0, 0, d)$, subject to the boundary conditions:

1. $V = 0$ when $z = 0$ (since the conducting plane is grounded), and
2. $V \rightarrow 0$ far from the charge (that is, for $x^2 + y^2 + z^2 \gg d^2$).

At $z > 0$, $V \neq \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ because

- q will induce a certain amount of negative charge on the nearby surface of the conductor; and therefore
- the total potential is due in part to q directly, and in part to this induced charge.

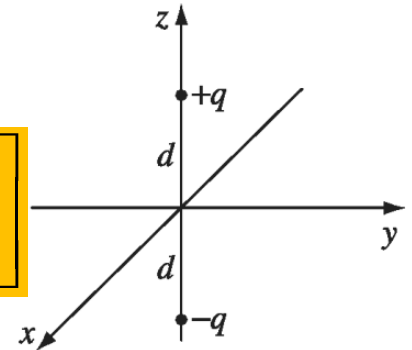


How can we calculate the potential, when we don't know how much charge is induced or how it is distributed?

Consider a new configuration for the method of images, two point charges $+q$ at $(0, 0, d)$ and $-q$ at $(0, 0, -d)$ and no conducting plane as shown.

The potential $V(x, y, z)$ at any point $z > 0$ is then

$$V(x, y, z) = \frac{1}{4\pi\epsilon_0} \left[\frac{q}{\sqrt{x^2 + y^2 + (z-d)^2}} - \frac{q}{\sqrt{x^2 + y^2 + (z+d)^2}} \right]$$



The denominators represent the distances from (x, y, z) to the charges $+q$ and $-q$, respectively.). It follows from the solution that

1. $V = 0$ when $z = 0$, and
2. $V \rightarrow 0$ for $x^2 + y^2 + z^2 \gg d^2$.

The new configuration happens to produce exactly the same potential as the first configuration, in the "upper" region $z \geq 0$ and gives the correct solution.

The uniqueness theorem certifies that if the solution satisfies Poisson's equation in the region of interest, and assumes the correct value at the boundaries, then it must be right.

Is this solution correct for the region $z < 0$?

- Below the plane (where the conductor was located), the solution is incorrect.
- $\mathbf{E} = 0$ in a conductor whereas for a system of charges, it is not.
- This minor difficulty is resolved by using the solution above the plane.

General properties for the method of images in planar geometries:

- 1) The image is the negative of the source. The magnitude of the image is the same as the source.
- 2) The geometry is reflected in the constant potential surface as in a mirror.
- 3) The image and source are at the same distance from the mirror.
- 4) Multiple sources produce multiple images, again like reflection in a mirror.
- 5) Single or multiple point or distributed charges in front of multiple mirrors also produce multiple images.

Important: Rules (1) and (3) only apply to planar surfaces. The rest apply in curved surfaces as well.

Induced surface charge and force

Can we use this potential to find out the electric field, surface charge distribution, and the force? Yes, we can.

We know that $\sigma = -\epsilon_0 \frac{\partial V}{\partial n}$

where $\frac{\partial V}{\partial n}$ is the normal derivative of V at the surface.

The normal direction is the z -direction, so $\sigma = -\epsilon_0 \left. \frac{\partial V}{\partial n} \right|_{z=0}$

$$\frac{\partial V}{\partial z} = \frac{1}{4\pi\epsilon_0} \left\{ \frac{-q(z-d)}{[x^2 + y^2 + (z-d)^2]^{3/2}} + \frac{q(z+d)}{[x^2 + y^2 + (z+d)^2]^{3/2}} \right\}$$

$$\sigma(x, y) = \frac{-qd}{2\pi(x^2 + y^2 + d^2)^{3/2}}$$

As expected, the induced charge is negative (assuming q is positive) and greatest at $x = y = 0$.

The total induced charge $Q = \int \sigma da$ can be easily calculated in the cylindrical polar coordinates (r, φ) , with $r^2 = x^2 + y^2$ and $da = r dr d\varphi$.

$$\sigma(r) = \frac{-qd}{2\pi(r^2 + d^2)^{3/2}},$$

$$Q = \int_0^{2\pi} \int_0^\infty \frac{-qd}{2\pi(r^2 + d^2)^{3/2}} r dr d\phi = \left. \frac{qd}{\sqrt{r^2 + d^2}} \right|_0^\infty = -q.$$

The total charge induced on the plane is $-q$.

The charge q is attracted toward the plane, because of the negative induced charge.

Since the potential in the vicinity of q is the same as in the analog problem (the one with $+q$ and $-q$ but no conductor), so also is the field and, therefore, the force is given by

$$\mathbf{F} = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{(2d)^2} \hat{\mathbf{z}}.$$

This image force contributes in large measure to preventing electrons from leaving the surfaces of conductors and is associated with the "work function" of conducting materials.

The force can also be calculated by using the expression for the force per unit area

$$\mathbf{f} = \frac{1}{2\epsilon_0} \sigma^2 \hat{\mathbf{n}}$$

Energy

Energy, however, is not same for both cases.

For two point charges, without the conductor, energy as per general relation is

$$W = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{2d}.$$

For a single charge and conducting plane the energy is half of this

$$W = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}. \quad \text{Why half?}$$

In the first case both the upper region ($z > 0$) and the lower region ($z < 0$) contribute and by symmetry they contribute equally.

But in the second case only the upper region contains a nonzero field, and hence the energy is half as great.

One can also determine the energy by calculating the work required to bring q in from infinity.

$$W = \int_{\infty}^d \mathbf{F} \cdot d\mathbf{l} = \frac{1}{4\pi\epsilon_0} \int_{\infty}^d \frac{q^2}{4z^2} dz = \frac{1}{4\pi\epsilon_0} \left(-\frac{q^2}{4z} \right) \Big|_{\infty}^d = -\frac{1}{4\pi\epsilon_0} \frac{q^2}{4d}.$$

One can also understand the difference in the amount of work done as follows:

- As q is moved toward the conductor, the work is done only on q .
- The motion of q will continue to redistribute the induced charge over the conductor, but this doesn't result in any work being done on the induced charges since the whole conductor is at potential zero.
- By contrast, if both the charges are brought toward each (with no conductor), the work is done on both of them, and the total is (again) twice as great.

A point charge outside a grounded conducting sphere

A point charge q is situated a distance a from the center of a grounded conducting sphere of radius R .

Find the potential outside the sphere.

Consider a completely different configuration, consisting of the point charge q together with another point charge q' located at a distance b from the center of the conducting sphere such that $b < R$.

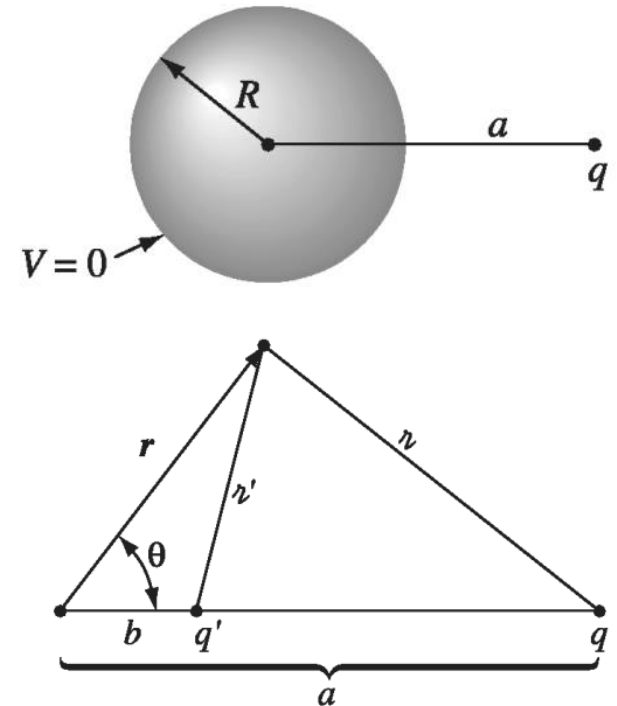
The potential at r is

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{r} + \frac{q'}{r'} \right)$$

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{r^2 + a^2 - 2ra \cos \theta}} + \frac{q'}{\sqrt{r^2 + b^2 - 2rb \cos \theta}} \right)$$

Now, at $\theta = 0$ and $\theta = \pi$, $V(r = R) = 0$, therefore we get

$$\frac{1}{4\pi\epsilon_0} \left(\frac{q'}{R - b} + \frac{q}{a - R} \right) = 0 \quad \text{and} \quad \frac{1}{4\pi\epsilon_0} \left(\frac{q'}{R + b} + \frac{q}{a + R} \right) = 0$$



Solving the two equations, we get

$$b = \frac{R^2}{a}, q' = \frac{-R}{a} q$$

- $|q'| < q$, and hence not all the electric field lines which begin at q terminate on the conducting sphere. The remainder go off to infinity.
- This “loss” of flux to infinity differs from the case of the conducting plane because the surface area of a sphere is finite.

The force of attraction between the charge and the sphere is

$$F = \frac{1}{4\pi\epsilon_0} \frac{qq'}{(a-b)^2} = -\frac{1}{4\pi\epsilon_0} \frac{q^2 Ra}{(a^2 - R^2)^2}.$$

What is $V(\mathbf{r})$, if the conducting sphere is maintained at a potential V_0 ?

Just add the potential from a second image with charge $q_0 = 4\pi\epsilon_0 R V_0$ placed at the center of the sphere (applying principle of superposition)

$$V(\mathbf{r}) = \frac{1}{4\pi\epsilon_0} \left(\frac{q}{\sqrt{r^2 + a^2 - 2ra \cos \theta}} + \frac{q'}{\sqrt{r^2 + b^2 - 2rb \cos \theta}} \right) + \frac{RV_0}{r}$$

Separation of variables

Laplace's equation can be solved directly, using the method of separation of variables, which is the physicist's favorite tool for solving partial differential equations.

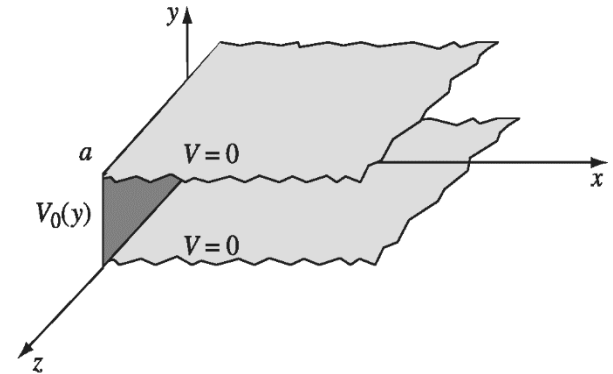
Applicability: The method is applicable in circumstances where the potential (V) or the charge density (σ) is specified on the boundaries of some region, and we are asked to find the potential in the interior (where $\rho = 0$).

Laplace's equation: $\nabla^2 V = 0$

Basic strategy: Look for solutions that are products of functions, each of which depends on only one of the coordinate, $V(x, y, z) = X(x)Y(y)Z(z)$.

Two infinite grounded metal plates

Two infinite grounded metal plates lie parallel to the xz plane, one at $y = 0$, the other at $y = a$. The left end, at $x = 0$, is closed off with an infinite strip insulated from the two plates and maintained at a specific potential $V_0(y)$. Find the potential inside this "slot."



The configuration is independent of z , so Laplace's equation reduces to two dimensions.

$$\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$$

The potential inside is subject to the **boundary conditions**

- (i) $V = 0$ when $y = 0$,
- (ii) $V = 0$ when $y = a$,
- (iii) $V = V_0(y)$ when $x = 0$,
- (iv) $V \rightarrow 0$ as $x \rightarrow \infty$.

Since the potential is specified on all boundaries, the answer is uniquely determined. **Note that (iv) is not mentioned explicitly but is necessary on physical grounds: the potential should drop to zero as we move away from $x = 0$.**

How to solve for the given configuration?

- Look for solutions in the form of products: $V(x,y) = X(x)Y(y)$.
- Substitute into Laplace's equation to obtain

$$\frac{1}{X} \frac{d^2 X}{dx^2} + \frac{1}{Y} \frac{d^2 Y}{dy^2} = 0.$$

The first term depends only on x and the second only on y .

- The sum of these two functions is zero, which implies these two functions must both be constant.

$$\frac{1}{X} \frac{d^2 X}{dx^2} = C_1 \quad \text{and} \quad \frac{1}{Y} \frac{d^2 Y}{dy^2} = C_2, \quad \text{with} \quad C_1 + C_2 = 0.$$

- One of these constants is positive, the other negative (or perhaps both are zero). In general, one must investigate all the possibilities; however, in this particular problem C_1 is positive and C_2 is negative. We write,

$$\frac{d^2 X}{dx^2} = k^2 X, \quad \frac{d^2 Y}{dy^2} = -k^2 Y.$$

A partial differential equation $\frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} = 0$ has been converted into two ordinary differential equations. **So what?**

Ordinary differential equations are a lot easier to solve.

➤ The solutions of the two equations are of the form

$$X(x) = Ae^{kx} + Be^{-kx}, \quad Y(y) = C \sin ky + D \cos ky$$

and the complete solution is written as

$$V(x, y) = (Ae^{kx} + Be^{-kx})(C \sin ky + D \cos ky)$$

➤ Let us now apply the boundary conditions to evaluate A , B , C , and D .

(iv) $\Rightarrow A = 0$.

$$V(x, y) = e^{-kx}(C \sin ky + D \cos ky) \quad (\text{Absorb } B \text{ in } C \text{ and } D).$$

(i) $\Rightarrow D = 0$.

$$V(x, y) = Ce^{-kx} \sin ky$$

(ii) $\Rightarrow \sin ka = 0$

$$k = \frac{n\pi}{a}, \quad n = 1, 2, 3, \dots \quad (n \neq 0, \text{ Why?})$$

At this point you can see why C_1 is positive and C_2 is negative:

If X were sinusoidal, we could never arrange for it to go to zero at infinity, and if Y were exponential we could not make it vanish at both 0 and a .

The final solution is of the form,

$$V(x, y) = C e^{-kx} \sin \frac{n\pi}{a} y$$

- $V(x, y)$ represents an infinite set of solutions (one for each n).
- None of them by itself satisfies the boundary condition (iii).

How to fix the boundary condition (iii)?

Utilize the Laplace's equation's linear property, in the sense that if $V_1, V_2, V_3, \dots$ satisfy it, so does any linear combination, $V = V_1\alpha_1 + V_2\alpha_2 + V_3\alpha_3 + \dots$ where $\alpha_1, \alpha_2, \alpha_3, \dots$ are arbitrary constants. Thus,

$$\nabla^2 V = \alpha_1 \nabla^2 V_1 + \alpha_2 \nabla^2 V_2 + \alpha_3 \nabla^2 V_3 + \dots = \alpha_1 \cdot 0 + \alpha_2 \cdot 0 + \alpha_3 \cdot 0 + \dots = 0$$

A Complete Solution in Fourier Series

Exploiting this fact, we put together the separable solutions to construct a much more general solution:

$$V(x, y) = \sum_{n=1}^{\infty} C_n e^{-n\pi x/a} \sin \frac{n\pi}{a} y$$

This still satisfies the first three boundary conditions; the question is, can we fit the last boundary condition? i.e.,

$$V(0, y) = \sum_{n=1}^{\infty} C_n \sin \frac{n\pi}{a} y = V_0(y)$$

This is a Fourier sine series. Virtually any function $V_0(y)$ can be expanded in such a series.

Use the so-called “Fourier’s trick” to find out the coefficients C_n , **and the trick is:**

Multiply the above equation by $\sin(n'\pi y/a)$ (where n' is a positive integer), and integrate from 0 to a :

$$\sum_{n=1}^{\infty} C_n \int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n'\pi y}{a}\right) dy = \int_0^a V_o(y) \sin\left(\frac{n'\pi y}{a}\right) dy$$

where,

$$\int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{n'\pi y}{a}\right) dy = \begin{cases} 0 & \text{if } n' \neq n \\ \frac{a}{2} & \text{if } n' = n \end{cases}$$

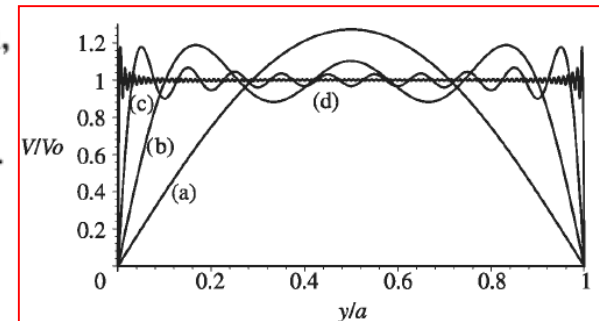
Now, all the terms in the series drop out, except where $n = n'$, and LHS is $\frac{a}{2} C_{n'}$.

$$C_n = \frac{a}{2} \int_0^a V_o(y) \sin\left(\frac{n\pi y}{a}\right) dy$$

Suppose the strip at $x = 0$ is a metal plate with constant potential V_0 (remember, it's insulated from the grounded plates at $y = 0$ and $y = a$). Then

$$C_n = \frac{2V_0}{a} \int_0^a \sin(n\pi y/a) dy = \frac{2V_0}{n\pi} (1 - \cos n\pi) = \begin{cases} 0, & \text{if } n \text{ is even,} \\ \frac{4V_0}{n\pi}, & \text{if } n \text{ is odd.} \end{cases}$$

$$\text{Therefore, } V(x, y) = \frac{4V_0}{\pi} \sum_{n=1,3,5,\dots} \frac{1}{n} e^{-n\pi x/a} \sin(n\pi y/a).$$



(a) $n=1$, (b) $n=5$, (c) $n=10$,
(d) $n=100$.

Questions?